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Informing an upper layer of barring alleviation for multiple access classes

Abstract

This document describes techniques and devices for informing an upper layer of barring alleviation for multiple access classes. In some situations, a radio resource control (RRC) layer of a user equipment (UE) can provide to an upper layer of the UE a barring message that indicates at least two access classes are barred in response to receiving a rejection message from a random access network (RAN). Responsive to the RRC layer determining that the at least two access classes are no longer barred, the RRC layer can provide the upper layer an alleviation message that indicates barring of the least two access classes is alleviated. By informing the upper layer of multiple access classes using the alleviation message, miscommunications regarding barring alleviation of the access classes can be avoided between the RRC layer and the upper layer.

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Background/Summary

RELATED APPLICATION(S) (1) This application is a national stage entry of International Application No. PCT/US2019/062948, filed Nov. 25, 2019, which claims the benefit of U.S. Provisional Application No. 62/773,123, filed Nov. 29, 2018, the disclosures which are incorporated herein by reference in their entirety.

BACKGROUND
(1) The evolution of wireless communication to 5th Generation (5G) standards and technologies provide higher data rates and greater capacity, with improved reliability and lower latency, which enhances mobile broadband services. 5G technologies also provide new classes of services for vehicular, fixed wireless broadband, and the Internet of Things (IoT). The specification of the features in the 5G air interface is defined as 5G New Radio (5G NR). To provide these services, the 5G standards enable a network to manage congestion based on access classes.
(2) In general, an access class indicates a purpose and priority associated with a user equipment (UE) connecting to the network. A higher-priority access class, for example, may be used for making emergency calls or may be assigned to a higher-priority user such as an emergency responder (e.g., a police officer or a firefighter). In contrast, a lower-priority access class may be used for non-emergency reporting of a remote humidity sensor's reading.
(3) If the network becomes congested, the network can deny a lower-priority UE's request to establish a connection using a lower-priority access class in order to allocate network resources to higher-priority UEs or other UEs establishing connections using higher-priority access classes. After a network denial of an establishment request, a set of lower-priority access classes (including the lower-priority access class of the establishment request) is considered barred at that UE for a predetermined amount of time. After the predetermined amount of time has passed, the UE can again attempt to use a lower-priority access class to connect to the network. In this manner, barring of the set of lower-priority access classes can be alleviated when congestion decreases.
(4) Sometimes, however, miscommunications can occur between different layers of a protocol stack implemented by a UE. These miscommunications can cause one of the layers to be unaware

that barring of an access class is alleviated. Consequently, the UE may be less likely to attempt to establish a connection having that access class, which can result in an unintended limitation of services that the UE may provide for a user.

SUMMARY

(5) Techniques and apparatuses are described for informing an upper layer of barring alleviation for multiple access classes. In some situations, a radio resource control (RRC) layer of a user equipment (UE) can provide to an upper layer of the UE a barring message that indicates a group of at least two access classes are barred. This can be in response to the RRC layer receiving a rejection message from a network that denies a request to establish a connection associated with a first access class of the group of at least two access classes. In this case, the rejection message causes multiple access classes to be barred as a group. Responsive to the RRC layer determining that the group of at least two access classes are no longer barred due to the rejection message, the RRC layer can provide the upper layer an alleviation message that indicates barring of the group of at least two access classes is alleviated. This can enable the upper layer to direct the RRC layer to attempt to establish a second connection associated with one access class from the group of at least two access classes (e.g., with the first access class or a second access class). By informing the upper layer of multiple access classes using the alleviation message, miscommunications regarding barring alleviation can be avoided between the RRC layer and the upper layer.

(6) Aspects described below include a method for informing an upper layer of a user equipment of barring alleviation for multiple access classes. As part of this method, the user equipment performs operations that include determining that a first access class is not barred at a radio resource control layer of the user equipment. The method also includes sending, by the radio resource control layer, a first connection request message to a network. The first connection request message requests permission to establish a first wireless connection associated with the first access class. The method further includes receiving a rejection message from the network at the radio resource control layer. The rejection message denies the request to establish the first wireless connection. Responsive to receiving the rejection message, the method includes providing, from the radio resource control layer to the upper layer, a first barring message that indicates barring of at least the first access class and a second access class. Responsive to receiving the rejection message, the method includes activating, at the radio resource control layer, a common bar timer. Responsive to the common bar timer becoming inactive, the method includes determining that the barring of at least the first access class and the second access class is alleviated based on respective specific bar timers associated with at least the first access class and the second access class being inactive. The method also includes providing, from the radio resource control layer to the upper layer, a first alleviation message that indicates alleviation of the barring of the first access class and the second access.

(7) Aspects described below also include a user equipment with a radio-frequency transceiver. The user equipment also includes a processor and memory system configured to perform any of the described methods.

(8) Aspects described below also include a system with means for informing an upper layer of barring alleviation for multiple access classes.

(9) A further aspect includes a processor-readable medium having instructions stored thereon that, when executed by a processor, cause the processor to perform the method.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Apparatuses of and techniques for informing an upper layer of barring alleviation for multiple access classes are described with reference to the following drawings. The same numbers are used throughout the drawings to reference like features and components:

(2) FIG. **1** illustrates an example wireless network environment in which informing an upper layer of barring alleviation for multiple access classes can be implemented.
(3) FIG. **2** illustrates an example device diagram of a user equipment and a base station for informing an upper layer of barring alleviation for multiple access classes.
(4) FIG. **3** illustrates an example block diagram of a wireless network stack model in which various aspects of informing an upper layer of barring alleviation for multiple access classes can be implemented.
(5) FIG. **4** illustrates details of example control transactions between entities for informing an upper layer of barring alleviation for multiple access classes.
(6) FIG. **5** illustrates details of example control transactions between entities informing an upper layer of barring alleviation for multiple access classes associated with different bar timers at a same time.
(7) FIG. **6** illustrates details of example control transactions between entities informing an upper layer of barring alleviation for multiple access classes associated with different bar timers at different times.
(8) FIG. **7** illustrates details of example control transactions between entities for informing an upper layer of barring alleviation for multiple access classes associated with different bar timers at a same time.
(9) FIG. **8** illustrates details of example control transactions between entities for informing an upper layer of barring alleviation for multiple access classes responsive to receiving approval to establish a connection.
(10) FIG. **9** illustrates details of example control transactions between entities for informing an upper layer of barring alleviation for multiple access classes responsive to selection of a different cell.
(11) FIG. **10** illustrates an example method for informing an upper layer of barring alleviation for multiple access classes.

DETAILED DESCRIPTION

Overview

(12) Techniques and apparatuses are described for informing an upper layer of barring alleviation for multiple access classes. Sometimes miscommunications can occur between different layers of a protocol stack implemented by a user equipment (UE), such as between a radio resource control (RRC) layer of the UE and an upper layer of the UE. These miscommunications can result in the upper layer being unaware that barring of an access class is alleviated. Consequently, the UE may be less likely to attempt to establish a connection associated with the previously barred access class, which can result in an unintended limitation of services that the UE may provide for a user.
(13) To address this issue, techniques and apparatuses are described for informing an upper layer of barring alleviation for multiple access classes. In some situations, the RRC layer can provide to the upper layer a barring message that indicates a group of at least two access classes are barred. This can be in response to the RRC layer receiving a rejection message from a network that denies a request to establish a connection associated with a first access class of the group of at least two access classes. In this case, the rejection message causes multiple access classes to be barred as a group. Responsive to the RRC layer determining that the group of at least two access classes are no longer barred due to the rejection message, the RRC layer can provide the upper layer an alleviation message that indicates barring of the group of at least two access classes is alleviated. This can enable the upper layer to direct the RRC layer to attempt to establish a second connection associated with one access class of the group of at least two access classes (e.g., with either the first access class or a second access class). By informing the upper layer of multiple access classes using the alleviation message, miscommunications regarding barring alleviation can be avoided between the RRC layer and the upper layer.
(14) Implementations in accordance with the present disclosure can have some, or all, of the

following advantages. The provision of a barring message, by the RRC layer to an upper layer, that indicates barring of multiple access classes can save computational power and battery power at the UE. For example, the barring message can prevent the upper layer from sending superfluous request messages to the RRC layer in respect of the second access class (and/or other access classes) after the network has rejected a request to establish a connection associated with the first access class. The provision of an alleviation message, by the RRC layer to an upper layer, that indicates alleviation of barring of multiple access classes can allow the upper layer to make better use of available radio resources following the alleviation. For example, the alleviation message can ensure that the upper layer is informed of each and every access class for which barring has been alleviated, thus avoiding miscommunications between the RRC layer and the upper layer that could otherwise cause the upper layer not to utilize available radio resources. Alternatively or in addition, the alleviation message can inform the upper layer when barring of a particular access class has been alleviated before expiry of a specific bar timer for that particular access class, thus allowing the upper layer to utilize newly-available radio resources. Other advantages will become apparent from the following description.

Example Environment

(15) FIG. **1** illustrates an example environment **100**, which includes a user equipment (UE) **110** that can communicate with a base station **120** that acts as a serving cell (serving base station **120**) through one or more wireless communication links **130** (wireless link **130**). For simplicity, the UE **110** is implemented as a smartphone but may be implemented as any suitable computing or electronic device, such as a mobile communication device, a modem, cellular phone, mobile phone, gaming device, navigation device, media device, laptop computer, desktop computer, tablet computer, smart appliance, vehicle-based communication system, or an Internet of Things (IoT) device, such as a sensor or an actuator.

(16) A macrocell, microcell, small cell, picocell, and the like, or any combination thereof may implement the base station **120** (e.g., an Evolved Universal Terrestrial Radio Access Network Node B, E-UTRAN Node B, evolved Node B, eNodeB, eNB, Next Generation eNB, ng-eNB, Next Generation Node B, gNode B, gNB, or the like), a base station transceiver system, a Wireless Local Access Network (WLAN) router, a satellite, a terrestrial television broadcast tower, an access point, a peer-to-peer device, another smartphone acting as a base station, and so forth.

(17) The base station **120** communicates with the UE **110** (e.g., UE **111** or UE **112**) using the wireless link **130** (e.g., wireless link **131** or wireless link **135**), which may be implemented as any suitable type of wireless link. The wireless links **130** include control and data communication, such as downlink of data and control information communicated from the base station **120** to the UE **110**, uplink of other data and control information communicated from the UE **110** to the base station **120**, or both. The wireless links **130** may include one or more wireless links (e.g., radio links) or bearers implemented using any suitable communication protocol or standard, or combination of communication protocols or standards such as 3.sup.rd Generation Partnership Project Long-Term Evolution (3GPP LTE), 5G NR, and so forth. Multiple wireless links **130** may be aggregated in a carrier aggregation to provide a higher data rate for the UE **110**. Multiple wireless links **130** from multiple base stations **120** may be configured for Coordinated Multipoint (CoMP) communication with the UE **110**. Additionally, the multiple wireless links **130** may be configured for single-RAT dual connectivity or multi-RAT dual connectivity (MR-DC).

(18) In FIG. **1**, the UE **110** can communicate with one or more base stations **124** (illustrated as base stations **121** and **122**) through one or more wireless links **130** (illustrated as wireless links **131** and **133**). The wireless links **131** and **133** may be implemented using a same communication protocol or communication standard, or different communication protocols or communication standards. Sometimes the UE **110** may communicate with another base station **125** (neighbor base station **125**) using a wireless link **135**. The wireless link **135** may be implemented using the same communication protocol or communication standard, or a different communication protocol or

communication standard, as the wireless link **131**.
(19) The base stations **120** (e.g., the base stations **121**, **122**, and **125**, and any additional base stations not illustrated for clarity) are collectively a Radio Access Network **140** (e.g., RAN, Evolved Universal Terrestrial Radio Access Network, E-UTRAN, 5G NR RAN, or NR RAN). The base stations **121** and **125** in the RAN **140** are connected to a 5.sup.th Generation Core network **150** (5GC **150**).
(20) The base stations **121** and **125** connect, at **102** and **104** respectively, to the 5GC **150** through an NG2 interface for control-plane signaling and using an NG3 interface for user-plane data communications. In addition to connecting with the 5GC **150**, the base stations **120** may communicate with each other. For instance, the base station **124** (e.g., base station **121**) and the neighbor base station **125** can communicate using an Xn Application Protocol (XnAP), at **116**, to exchange user-plane and control-plane data. Additionally, the base stations **121** and **122** can communicate using an Xn interface, or an X2 interface, at **117**, to exchange user-plane and control-plane data.
(21) The 5GC **150** includes an Access and Mobility Management Function **152** (AMF **152**), which provides control-plane functions such as registration and authentication of multiple UE devices **110**, authorization, and mobility management in the 5G NR network. The AMF **152** communicates with the base stations **120** in the RAN **140** and also communicates with multiple UEs **110**, using the base stations **120**.
Example Devices
(22) FIG. **2** illustrates an example device diagram **200** of the UE **110** and the base station **120**. The UE **110** and the base station **120** may include additional functions and interfaces omitted from FIG. **2** for the sake of clarity. The UE **110** includes antennas **202**, a radio-frequency (RF) front end **204**, an LTE transceiver **206**, and a 5G NR transceiver **208** for communicating with one or more base stations **120** in the RAN **140**. The RF front end **204** can couple or connect the LTE transceiver **206** and the 5G NR transceiver **208** to the antennas **202** to facilitate various types of wireless communication. The antennas **202** may include an array of multiple antennas that are configured similar to or different from each other. The antennas **202** and the RF front end **204** can be tuned to, and/or be tunable to, one or more frequency bands defined by the 3GPP LTE and 5G NR communication standards and implemented by the LTE transceiver **206** and/or the 5G NR transceiver **208**. Additionally, the antennas **202**, the RF front end **204**, the LTE transceiver **206**, and/or the 5G NR transceiver **208** may be configured to support beamforming for the transmission and reception of signals with the base station **120**. By way of example and not limitation, the antennas **202** and the RF front end **204** can be implemented for operation in sub-gigahertz bands, sub-6 GHz bands, and/or above 6 GHz bands that are defined by the 3GPP LTE and 5G NR communication standards.
(23) The UE **110** also includes processor(s) **210** and computer-readable storage media **212** (CRM **212**). The processor **210** may be a single-core processor or a multi-core processor composed of a variety of materials, such as silicon, polysilicon, high-K dielectric, copper, and so on. The computer-readable storage media described herein excludes propagating signals. CRM **212** may include any suitable memory or storage device such as random-access memory (RAM), static RAM (SRAM), dynamic RAM (DRAM), non-volatile RAM (NVRAM), read-only memory (ROM), or Flash memory useable to store device data **214** of the UE **110**. The device data **214** includes user data, multimedia data, beamforming codebooks, applications, and/or an operating system of the UE **110**, which are executable by the processor **210** to enable user-plane communication, control-plane signaling, and user interaction with the UE **110**.
(24) The device diagram for the base station **120**, shown in FIG. **2**, includes a single network node (e.g., a gNB). The functionality of the base station **120** may be distributed across multiple network nodes or devices and may be distributed in any fashion suitable to perform the functions described herein. The base station **120** includes antennas **252**, a radio-frequency (RF) front end **254**, one or

more LTE transceivers **256**, and/or one or more 5G NR transceivers **258** for communicating with the UE **110**. The RF front end **254** of the base station **120** can couple or connect the LTE transceivers **256** and the 5G NR transceivers **258** to the antennas **252** to facilitate various types of wireless communication. The antennas **252** may include an array of multiple antennas that are configured similar to or different from each other. The antennas **252** and the RF front end **254** can be tuned to, and/or be tunable to, one or more frequency bands defined by the 3GPP LTE and 5G NR communication standards, and implemented by the LTE transceivers **256**, and/or the 5G NR transceivers **258**. Additionally, the antennas **252**, the RF front end **254**, the LTE transceivers **256**, and/or the 5G NR transceivers **258** may be configured to support beamforming, such as Massive-MIMO, for the transmission and reception of communications with the UE **110**.
(25) The base station **120** also includes processor(s) **260** and computer-readable storage media **262** (CRM **262**). The processor **260** may be a single-core processor or a multi-core processor composed of a variety of materials, such as silicon, polysilicon, high-K dielectric, copper, and so on. CRM **262** may include any suitable memory or storage device such as random-access memory (RAM), static RAM (SRAM), dynamic RAM (DRAM), non-volatile RAM (NVRAM), read-only memory (ROM), or Flash memory useable to store device data **264** of the base station **120**. The device data **264** includes network scheduling data, radio resource management data, beamforming codebooks, applications, and/or an operating system of the base station **120**, which are executable by the processor **260** to enable communication with the UE **110**.
(26) The CRM **262** also includes a base station manager **266**. Alternatively or additionally, the base station manager **266** may be implemented in whole or part as hardware logic or circuitry integrated with or separate from other components of the base station **120**. In at least some aspects, the base station manager **266** configures the LTE transceivers **256** and the 5G NR transceivers **258** for communication with the UE **110**, as well as communication with the 5GC **150**. The base station **120** includes an inter-base station interface **268**, such as an Xn and/or X2 interface, to exchange user-plane and control-plane data with another base station **120** and manage communications between the base stations **120** with the UE **110**. The base station **120** also includes a core network interface **270** to exchange information with core network functions and entities, such as those associated with the 5GC **150**.
User Plane and Control Plane Signaling
(27) FIG. **3** illustrates an example block diagram **300** of a wireless network stack model **300** that characterizes a communication system for the example environment **100**, in which various aspects of informing an upper layer of barring alleviation for multiple access classes can be implemented. The wireless network stack **300** includes a user plane **302** and a control plane **304**. Upper layers of the user plane **302** and the control plane **304** share common lower layers in the wireless network stack **300**. Wireless devices, such as the UE **110** or the base station **120**, implement each layer as an entity for communication with another device using the protocols defined for the layer. For example, a UE **110** uses a Packet Data Convergence Protocol (PDCP) entity to communicate to a peer PDCP entity in a base station **120** using the PDCP.
(28) The shared lower layers include a physical (PHY) layer **306**, a Media Access Control (MAC) layer **308**, a Radio Link Control (RLC) layer **310**, and a PDCP layer **312**. The PHY layer **306** provides hardware specifications for devices that communicate with each other. As such, the PHY layer **306** establishes how devices connect to each other, assists in managing how communication resources are shared among devices, and the like.
(29) The MAC layer **308** specifies how data is transferred between devices. Generally, the MAC layer **308** provides a way in which data packets being transmitted are encoded and decoded into bits as part of a transmission protocol.
(30) The RLC layer **310** provides data transfer services to higher layers in the wireless network stack **300**. Generally, the RLC layer **310** provides error correction, packet segmentation and reassembly, and management of data transfers in various modes, such as acknowledged,

unacknowledged, or transparent modes.

(31) The PDCP layer **312** provides data transfer services to higher layers in the wireless network stack **300**. Generally, the PDCP layer **312** provides transfer of user plane **302** and control plane **304** data, header compression, ciphering, and integrity protection.

(32) Above the PDCP layer **312**, the wireless network stack splits into the user-plane **302** and the control-plane **304**. Layers of the user plane **302** include an optional Service Data Adaptation Protocol (SDAP) layer **314**, an Internet Protocol (IP) layer **316**, a Transmission Control Protocol/User Datagram Protocol (TCP/UDP) layer **318**, and an application layer **320** that transfers data using the wireless link **106**. The optional SDAP layer **314** is present in 5G NR networks. The SDAP layer **314** maps a Quality of Service (QoS) flow for each data radio bearer and marks QoS flow identifiers in uplink and downlink data packets for each packet data session. The IP layer **316** specifies how the data from the application layer **320** is transferred to a destination node. The TCP/UDP layer **318** is used to verify that data packets intended to be transferred to the destination node reached the destination node, using either TCP or UDP for data transfers by the application layer **320**. In some implementations, the user plane **302** may also include a data services layer (not shown) that provides data transport services to transport application data, such as IP packets including web-browsing content, video content, image content, audio content, social media content, and so forth.

(33) The control plane **304** includes a Radio Resource Control (RRC) layer **324** and a Non-Access Stratum (NAS) layer **326**. The RRC layer **324** establishes and releases connections and radio bearers, broadcasts system information, performs power control, and so forth. The RRC layer **324** also controls a resource control state of the UE **110** and directs the UE **110** to perform operations according to the resource control state. Example resource control states can include a connected state (e.g., an RRC connected state) or a disconnected state, such as an inactive state (e.g., an RRC inactive state) or an idle state (e.g., an RRC idle state). In general, if the UE **110** is in the connected state, the connection with the base station **120** is active. In the inactive state, the connection with the base station **120** is suspended. If the UE **110** is in the idle state, the connection with the base station **120** is released. Generally, the RRC layer **324** supports 3GPP access but does not support non-3GPP access (e.g., Wi-Fi).

(34) The NAS layer **326** provides support for mobility management (e.g., using a 5.sup.th Generation Mobility Management (5GMM) layer **328**) and packet data bearer contexts (e.g., using a 5.sup.th Generation Session Management (5GSM) layer **330**) between the UE **110** and entities or functions in the core network, such as the Access and Mobility Management Function **152** (AMF **152**) of the 5GC **150** or the like. The NAS layer **326** supports both 3GPP access and non-3GPP access.

(35) In the UE **110**, each layer in both the user plane **302** and the control plane **304** of the wireless network stack **300** interacts with a corresponding peer layer or entity in the base station **120**, a core network entity or function, and/or a remote service, to support user applications and control operation of the UE **110** in the RAN **140**.

Informing an Upper Layer of Barring Alleviation for Multiple Access Classes

(36) FIG. **4** illustrates details of example control transactions between entities for informing an upper layer of a UE of barring alleviation at that UE for multiple access classes. In this example, communications are transmitted between various layers of the communication protocol stack described in FIG. **3** of a UE **110** and a RAN **140**. For clarity, only a few layers of the wireless protocol stack of the UE **110** are illustrated, although more layers may be involved to efficiently process messages. In FIG. **4**, example protocol layers in the UE **110** include a UE upper layer **402** and a UE RRC layer **404**. The UE upper layer **402** can include the NAS layer **326** or the 5GMM layer **328** of FIG. **3**. The UE **110** uses these protocol layers to communicate with corresponding protocol layers in the RAN **140**. More generally, the term “upper layer” as used herein may refer to any layer of a wireless network stack that is above (or higher than) the UE RRC Layer **404**.

(37) At **405**, optional transactions identified by A**1** through A**2** can occur as further described in FIG. **5**. At **410**, the UE **110** is in a disconnected RRC state, such as the inactive state or the idle state. While in the disconnected RRC state, the UE **110** may attempt to establish (or re-establish) a connection with the RAN **140** using a particular access class (AC).

(38) Each access class is associated with an access category type and has an access category number. The access category type can include a standardized access category type, such as a mobile-terminated (MT) access type, a delay-tolerant type, an emergency type, a mobile-originated (MO) signal type, a MO voice call (MO MMTel Voice) type, a MO video call (MO MMTel Video) type, a MO Short Message Service (SMS) and SMS over Internet Protocol (SMSoIP) type, a MO data type, or a RAN area update type. Other access category types can include an operator-defined access category type specified by the RAN **140**. Example operator-defined access category types can include a data network name (DNN) type, a 5G QoS Identifier (5QI) type, an operating system (OS) identity (ID) or an OS application (App) ID of an application triggering the access attempt, a single network slice selection assistance information (S-NSSAI) type, or a standardized access category type as described above.

(39) Access category numbers can include one number from a range of numbers, such a number between zero and sixty-three. In some cases, different groups of access category numbers can be available to different entities. For example, access category numbers between zero and fifteen can be available to the UE **110**, and access category numbers between thirty-two and sixty-three can be reserved for use by the RAN **140**. In this example, access category numbers sixteen to thirty-one can be undefined or reserved for future use.

(40) Access classes can also be categorized as non-exempt or exempt. In general, the RAN **140** can bar a non-exempt access class and cannot bar an exempt access class. A non-exempt access class may be considered barred if the RAN **140** informs the UE RRC layer **404** that connections using the non-exempt access class are temporarily unsupported. The RAN **140**, for instance, can directly reject a request to establish a connection associated with the non-exempt access class to direct the UE RRC layer **404** to consider the non-exempt access class as temporarily barred. Generally, the RAN **140** may temporarily bar lower-priority UEs using non-exempt access classes (e.g., lower-priority access classes) and may not bar lower-priority UEs using exempt access classes (e.g., higher-priority access classes) during situations of high network traffic.

(41) Example types of exempt access classes can include an exempt access class that enables paging of the UE **110**, such as an access class associated with the MT access type and having an access category number of zero (referred to herein as AC**0**). Another example exempt access class that enables a user to make an emergency call can be associated with the emergency type and have an access category number of two (referred to herein as AC**2**). Example types of non-exempt access classes can include an access class associated with the MO signal type and having an access category number of three (referred to herein as AC**3**), another access class associated with the MO MMTel Voice type and having an access category number of four (referred to herein as AC**4**), an additional access class associated with the MO SMS and SMSoIP type and having an access category number of six (referred to herein as AC**6**), an access class associated with the MO data type and having an access category number of seven (referred to herein as AC**7**), another access class associated with the RAN area update type and having an access category number of eight (referred to herein as AC**8**), and the like. Sometimes a subscriber identification module (SIM) card of the UE **110** may include a list of exempt and/or non-exempt access classes that the UE **110** can use.

(42) At **420**, the UE RRC layer **404** performs a unified access control (UAC) procedure, defined by the 3GPP standard TS 38.331 specification, to determine if a first access class **422** (first AC **422**) is barred or is not barred. This determination can be based on system information provided by the RAN **140** to the UE **110**, such as information provided using a system information block (SIB) (e.g., SIB**1**). In particular, the system information can provide a barring threshold that the UAC

procedure uses to determine whether or not the first access class **422** is barred. In general, the barring threshold affects a probability of the UE RRC layer **404** determining that the first access class **422** is barred. Using the system information, the RAN **140** can inform multiple UEs **110** of the barring thresholds for multiple access classes. In some temporal situations during which network traffic increases, the RAN **140** can update the system information to adjust the barring thresholds to increase a likelihood that one or more of the access classes are determined to be barred by any particular UE's RRC layer.

(43) In some aspects, the UAC procedure can be performed in response to a decision by the UE RRC layer **404** to attempt to connect to the RAN **140**. If the UE RRC layer **404** decides to perform a RAN area update procedure, for example, the first access class **422** can be AC**8**. As such, the UE RRC layer **404** performs the UAC procedure at **420** to determine if AC**8** is barred.

(44) In other aspects, the UE upper layer **402** can send (e.g., provide) a request message **416** to the UE RRC layer **404**, as shown at **415**. This can occur if a registration timer maintained by the UE upper layer **402** expires and the UE upper layer **402** has a MO message to send to the RAN **140**. In this case, the first access class **422** can be AC**3** and the UE RRC layer **404** can determine if AC**3** is barred using the UAC procedure. In general, the request message **416** directs the UE RRC Layer **404** to attempt to establish a connection associated with the first access class **422** based on a UAC procedure determination by the UE RRC layer **404** that the first access class **422** is not barred.

(45) In still other cases, the first access class **422** can be AC**4** if the user attempts to make a voice call, AC**6** if the user attempts to make a video call, or AC**7** if the user attempts to send a text message. In FIG. **4**, the UE RRC layer **404** determines that the first access class **422** is not barred using the UAC procedure and proceeds to attempt to connect to the RAN **140**.

(46) At **425**, the UE RRC layer **404** transmits (e.g., sends or communicates) a connection request message **426** to the RAN **140**. The connection request message **426** can be an RRC setup request message or an RRC resume request message (e.g., an RRCResumeRequest message or an RRCConnectionResumeRequest message). At this time, for this example, the network congestion is high, and even though the RAN **140** may have adjusted the barring threshold in the system information block to alleviate congestion, the UE RRC layer **404**'s UAC procedure at **420** determined that the first access class **422** is not barred. To further alleviate congestion, the RAN **140** can respond to the connect request message **426** by sending a rejection message **432** to the UE RRC layer **404**. This rejection message **432** enables the RAN **140** to control access based on a UE **110**'s priority relative to other UEs. For instance, the RAN **140** can accept requests from higher-priority UEs **110** and reject requests from lower-priority UEs **110** or UEs requesting connections having lower priority access classes.

(47) At **430**, the UE RRC layer **404** receives the rejection message **432** from the RAN **140** that denies the UE RRC layer **404**'s request to establish a connection. The rejection message **432** can be an RRCReject message. The rejection message **432** enables the RAN **140** to manage network congestion and ensure that network resources are available for exempt access classes or higher-priority UEs **110**. Generally, the rejection message **432** directs the UE RRC layer **404** to determine that a set of at least two access classes are barred temporarily. The at least two access classes can include the first access class **422** and a second access class **444** (second AC **444**), both of which can be non-exempt access classes. Additionally, the rejection message **432** can include a wait time **434**, which specifies a duration for which the UE RRC layer **404** determines that the at least two access classes are barred. In effect, the wait time **434** directs the UE RRC layer **404** to wait a determined amount of time before attempting to connect to the RAN **140** with any of the barred access classes.

(48) Responsive to receiving the rejection message **432**, the UE RRC layer **404** starts (e.g., initiates) a common bar timer at **435** with the wait time **434** and sends a barring message **442** to the UE upper layer **402** at **440**. For drawing clarity reasons, the starting of the common bar timer is shown to occur before the sending of the barring message **442**; alternatively, the barring message **442** can be sent before the common bar timer starts, or both can occur at the same time. In general,

the UE RRC layer **404** uses the common bar timer to indicate a duration for which multiple access classes are barred at the UE. An example of a common bar timer is timer T**302**. Timer T**302** is described by the 3GPP standard TS 38.331 specification. The barring message **442** indicates to the upper layer **402** that the at least two access classes (e.g., the first access class **422** and the second access class **444**) are barred so that the upper layer **402** does not provide a request message for a barred access class (e.g., similar to message **416**) to the UE RRC layer **404** until after the bar is alleviated. Pausing upper layer **402** request messages during a barred duration saves computational power at the RRC layer **404**. This, in turn, can conserve power (e.g., battery power) at the UE **110**. Accordingly, the common bar timer in element **435** is associated with the access classes identified in the barring message **442**. As an example, the first access class **422** can be AC**3** and the second access class **444** can be AC**6**. The UE RRC layer **404** can also not include some access classes, such as the exempt access classes AC**0** and/or AC**2**, in the barring message **442**. In this manner, the UE RRC layer **404** can inform the UE upper layer **402** that some access classes are barred (e.g., those that are listed in the barring message **442**) and can inform the UE upper layer **402** that other access classes are not barred (e.g., those that are not listed in the barring message **442**).

(49) For simplicity, the transactions described between the optional sending of the request message **416** at **415** and the sending of the barring message **442** at **440** are generally identified by B. At **445**, other optional transactions identified by C can occur, as further described in FIGS. **5**, **8**, and **9**.

(50) In some situations, the common bar timer becomes inactive (e.g., expires or stops) at **450**. Responsive to the common bar timer becoming inactive, the UE RRC layer **404** sends an alleviation message **456** to the UE upper layer **402** at **455**. The alleviation message **456** indicates that barring of the group of access classes (such as the first access class **422** and the second access class **444**) are alleviated. This can re-enable the UE upper layer **402** to send a second request message associated with one of the previously-barred access classes (e.g., with either the first access class **422** or the second access class **444**).

(51) In some aspects, the UE RRC layer **404** can determine if at least one specific bar timer (to be described in association with FIG. **5**) associated with at least one of the previously barred access classes is active or running. If the specific bar timer is not running (e.g., inactive) or had previously expired (as will be described in further detail in FIG. **5**), the UE RRC layer **404** can indicate using the alleviation message **456** that barring of the access class associated with the specific bar timer is alleviated. By informing the UE upper layer **402** of the alleviation of multiple access classes that were previously barred (e.g., according to the barring message **442** or both barring messages **442** and **522**), miscommunications regarding barring alleviation can be avoided between the UE RRC layer **404** and the UE upper layer **402**.

(52) In general, the transactions that occur between elements **450** and **455** are identified by D. Some variations within D can occur, as further described with respect to FIGS. **5** and **7-9**. At **460**, other optional transactions identified by E can occur, as further described in FIG. **6**.

(53) At **465**, after the UE upper layer **402** received the alleviation message **456** re-enabling the upper layer to send request messages associated with a previously-barred access class, the UE upper layer **402** sends a second request message **466** to the UE RRC layer **404** to direct the UE **110** to attempt to establish a second connection associated with either the first access class **422** or the second access class **444**. This causes the UE RRC layer **404** to perform a second UAC procedure and send a second connection request message if the RRC layer determines that the access class is not barred (not shown but similar to elements **420** and **425** above).

(54) FIG. **5** illustrates details of example control transactions between entities for informing an upper layer of barring alleviation for multiple sets of access classes associated with different bar timers running during overlapping time periods. For simplicity, similar transactions previously described with respect to FIG. **4** are not shown explicitly. Instead, these transactions or operations are generically referenced using the identifiers A**1** through F. In FIG. **5**, the UE RRC layer **404** informs the UE upper layer **402** of multiple sets of access classes that are barred using multiple

barring messages (e.g., using barring messages **442** and **522**). Accordingly, the UE RRC layer **404** maintains multiple bar timers that are associated with the different sets of access classes. In this example, the multiple bar timers include the common bar timer of FIG. **4**, which is associated with at least two access classes, and a specific bar timer, which is associated with a particular access class. In some situations, the third access class **508** can be a different access class from the at least two access classes (e.g., the first access class **422** and the second access class **444**) discussed with respect to FIG. **4**. In other situations, the third access class **508** and the second access class **444** can be a same access class.

(55) Starting below the dashed horizontal line, at **505**, the UE upper layer **402** sends a request message **506** that directs the UE RRC layer **404** to attempt to establish a connection associated with a third access class **508**. At **510**, the UE RRC layer **404** performs the UAC procedure at **510**, similar to the UAC procedure performed at **420** in FIG. **4**. In this case, however, the UAC procedure determines that the third access class **508** is barred.

(56) Responsive to the determination that the third access class **508** is barred, the UE RRC layer **404** starts the specific bar timer with a second wait time **516** at **515** and sends a second barring message **522** at **520** to inform the UE upper layer **402** that the third access class **508** is barred. As an example, the specific bar timer can be timer T**390** or timer T**309**. Timer T**390** is described by the 3GPP standard TS 38.331 specification. The UE RRC layer **404** can determine the second wait time **516**. In this case, a value of the second wait time **516** causes the specific bar timer to expire before the common bar timer expires.

(57) In some cases, the third access class **508** can be an access class available to the UE **110**, such as AC**3**. In other cases, the third access class **508** can be an access class that is reserved for the RAN **140**. If the access class is reserved for the network, the RAN **140** can inform the UE upper layer **402** of the availability of this access class while the UE **110** is in a connected state, as described at the top of FIG. **5**.

(58) Prior to **505**, the UE **110** is in a connected RRC state at **525**. At **530**, the RAN **140** sends an update message **532** (e.g., a configuration update command) to the UE upper layer **402** to inform the UE upper layer **402** of the third access class **508**. As an example, the third access class **508** can be associated with the MO data type and have an access category number of thirty-seven (referred to herein as AC**37**). In some situations, the first access class **422** and the third access class **508** can be associated with a same access category type and have different access category numbers. At **535**, the UE **110** transitions from the connected RRC state to the disconnected RRC state and operations continue as described above at A**2** in FIG. **5**.

(59) After the RRC layer sends the barring message **522** at **520**, other operations as identified by B in FIG. **4** can occur at **540**. During this time, the UE RRC layer **404** receives the rejection message **432** from the RAN **140**, starts the common bar timer in element **435**, and sends the barring message **442** to the UE upper layer **402**, as described in FIG. **4**.

(60) At **545**, the specific bar timer expires before or during an expiration of the common bar timer at **450**. Similar to FIG. **4**, the UE RRC layer **404** sends the alleviation message **456** to the UE upper layer **402** at **550**. In contrast to FIG. **4**, however, the alleviation message **456** of FIG. **5** indicates that barring of the third access class **508** is alleviated in addition to the barring alleviation of the group of at least two access classes (e.g., the first access class **422** and the second access class **444**). Although not explicitly shown, the UE **110** can perform operations identified by E and F in FIG. **4** after the UE RRC layer **404** sends the alleviation message **456** at **550**. By informing the UE upper layer **402** of the alleviation of multiple sets of access classes that were previously barred according to multiple barring messages **442** (of FIG. **4**) and **522** (of FIG. **5**), miscommunications regarding barring alleviation can be avoided between the UE RRC layer **404** and the UE upper layer **402**.

(61) In a different situation not shown in FIG. **5**, the common bar timer may not be running at **545** (e.g., the operations identified by B at **540** did not occur). In this case, the UE RRC layer **404** can determine that the common bar timer is not running and send an alleviation message to the UE

upper layer **402** indicating that barring of the third access class **508** is alleviated in response to the specific bar timer expiring at **545**.

(62) FIG. **6** illustrates details of example control transactions between entities for informing an upper layer of barring alleviation for multiple sets of access classes associated with different bar timers that expire at different times. For simplicity, similar transactions previously described with respect to FIGS. **4** and **5** are not shown explicitly. Instead, these transactions or operations are generically referenced using the identifiers A**1** through F. In FIG. **6**, the UE RRC layer **404** informs the UE upper layer **402** of multiple sets of access classes that are barred using multiple barring messages (e.g., using barring messages **442** and **522**). Accordingly, the UE RRC layer **404** maintains multiple bar timers that are associated with the different groups of access classes. In this example, the multiple bar timers include the common bar timer of FIG. **4**, which is associated with a set of access classes containing at least two access classes, and the specific bar timer of FIG. **5**, which is associated with the third access class **508**. In contrast to FIG. **5**, the specific bar timer of FIG. **6** expires after the common bar timer. Similar to FIG. **5**, the third access class **508** can be an access class available to the UE **110** (e.g., AC**3**) or an access class reserved for the RAN **140** (e.g., AC**37**). In some situations, the third access class **508** can be a different access class than the first access class **422** and the second access class **444**. In other situations, the third access class **508** and the second access class **444** can be a same access class.

(63) At **605**, the operations identified by A**2** or the operations identified by both A**1** and A**2** occur, as described above with respect to FIG. **5**. As such, the UE RRC layer **404** informs the UE upper layer **402** that the third access class **508** is barred. In this case, a value of the second wait time **516** causes the specific bar timer to expire after the common bar timer expires. At **610**, the operations identified by B through D occur as described above with respect to FIG. **4**. During this time, the common bar timer expires and UE RRC layer **404** sends the alleviation message **456** to the UE upper layer **402** to indicate alleviation of the barring of the at least two access classes (the first access class **422** and the second access class **444**).

(64) In this case, the barring of the third access class **508** is not alleviated using the alleviation message **456** of FIG. **4** (e.g., the third access class **508** is not included in the alleviation message **456**) because the specific bar timer for the third access class **508** is still running. Accordingly, the UE upper layer **402** considers the third access class **508** as barred during the operations at **610**.

(65) At **615**, the specific bar timer expires and the UE RRC layer **404** sends a second alleviation message **622** to the UE upper layer **402** at **620**. In this case, the second alleviation message **622** indicates that barring of the third access class **508** is alleviated. Although not explicitly shown, the UE can also perform operations identified by F in FIG. **4** after the RRC layer sends the alleviation message **622** at **620**. By sending multiple alleviation messages (e.g., the alleviation message **456** at **455** and the second alleviation message **622** at **620**), the UE RRC layer **404** can inform the UE upper layer **402** that different access classes are alleviated at different times, even if the access classes have similar access category types. For example, the UE RRC layer **404** can inform the UE upper layer **402** that the AC**7** is alleviated using the alleviation message **456** and can inform the UE upper layer **402** that the AC**37** is alleviated using the alleviation message **622**.

(66) FIG. **7** illustrates details of example control transactions between entities for informing an upper layer of barring alleviation for multiple groups of access classes associated with different bar timers running during overlapping time periods. For simplicity, similar transactions previously described with respect to FIGS. **4** and **5** are not shown explicitly and are generically referenced using identifiers A**1** through F. Similar to FIGS. **5** and **6**, the UE RRC layer **404** in FIG. **7** informs the UE upper layer **402** of multiple groups of access classes that are barred using multiple barring messages (via the barring message **522** of FIG. **5** and using the barring message **442** of FIG. **4**). Accordingly, the UE RRC layer **404** maintains multiple bar timers such as the common bar timer of FIG. **4** and the specific bar timer of FIG. **5**. Instead of sending the second alleviation message **622** upon expiration of the specific bar timer as shown in FIG. **6**, however, the UE RRC layer **404** of

FIG. **7** sends a single alleviation message **456** for the multiple sets of access classes. Similar to FIGS. **5** and **6**, the third access class **508** can be an access class available to the UE **110** (e.g., AC**3**) or an access class reserved for the RAN **140** (e.g., AC**37**). In some situations, the third access class **508** can be a different access class than the first access class **422** and the second access class **444**. In other situations, the third access class **508** and the second access class **444** can be a same access class.

(67) At **705**, operations identified by A**1** and A**2** can be performed as described with respect to FIG. **5**. In this case, a value of the second wait time **516** causes the specific bar timer to expire after the common bar timer expires. At **710**, operations identified by B and C can be performed as described with respect to FIG. **4**.

(68) As part of the operations identified by D, the common bar timer expires prior to an expiration of the specific bar timer at **715**. Responsive to the common bar timer expiring, the UE RRC layer **404** stops the specific bar timer at **720** and sends the alleviation message **456** to the UE upper layer **402** at **725**. In this example, the alleviation message **456** indicates alleviation of the barring of both sets of access classes (the first access class **422**, the second access class **444**, and the third access class **508**). In this manner, the third access class **508** can be alleviated prior to an expiration of the specific bar timer when the specific bar timer is set to expire after the common bar timer. In FIG. **7**, a single alleviation message **456** informs the UE upper layer **402** of barring alleviation for multiple sets of access classes that are associated with different bar timers responsive to the common bar timer expiring before the specific bar timer expires. Although not explicitly shown, the UE can also perform operations identified by E and F in FIGS. **4** and **6** after the RRC layer sends alleviation message **456** at **725**.

(69) FIG. **8** illustrates details of example control transactions between entities for informing an upper layer of barring alleviation for multiple access classes responsive to receiving approval to establish a connection. For simplicity, similar transactions that are described above with respect to FIG. **4-7** are not shown explicitly. Instead, these transactions or operations are generically referenced using the identifiers A**1** through F. In this situation, an operation within the transactions identified by C direct the UE RRC layer **404** to stop the common bar timer and inform the UE upper layer **402** of barring alleviation for at least two access classes.

(70) At **805** and **810**, operations identified by A**1** and A**2** in FIG. **5** and operations identified by B in FIG. **4** can be performed. During the operations of B, the RRC layer starts a common bar timer as shown by element **435** in FIG. **4**. At **815**, the UE RRC layer **404** receives a request message **816** indicating a fourth access class **818** that is not included in the set of at least two access classes in a barring message **442**, which directs the UE RRC layer to attempt to establish a connection with the RAN **140**.

(71) At **820**, the UE RRC layer **404** performs the UAC procedure, which determines that the fourth access class **818** is not barred. As an example, the fourth access class **818** can be an exempt access class, such as AC**0** or AC**2**. Based on the determination that the fourth access class **818** is not barred, the UE RRC layer **404** proceeds to attempt to establish the connection with the RAN **140** by sending a connection request message **826** at **825**.

(72) At **830**, the UE RRC layer **404** receives a grant message **832** from the RAN **140** that approves the UE RRC layer **404**'s request to establish the connection using the fourth access class **818**. The grant message **832** can be an RRC setup message or an RRC resume message.

(73) At **835**, responsive to receiving the grant message **832**, the UE RRC layer **404** stops the common bar timer, which was previously started at **810** in response to the UE RRC layer **404** receiving the rejection message **432** as shown in FIG. **4** at element **435**. The UE RRC layer **404** also sends the alleviation message **456** to the UE upper layer **402** at **845**. The alleviation message **456** indicates that barring of the set of at least two access classes (e.g., the first access class **422** and the second access class **444**) is alleviated. If a specific bar timer was started at **805** (as shown in FIG. **5** element **515**), the specific bar timer can also be stopped at **840** in response to the UE RRC

layer **404** receiving the grant message **832**. Instead of waiting for the common bar timer to expire as described in FIGS. **4-7**, the operations of FIG. **8** quickly inform the UE upper layer **402** that the access classes that were previously barred are alleviated responsive to the UE RRC layer **404** receiving the grant message **832**. Although not explicitly shown, the UE can also perform operations identified by E and F in FIGS. **4** and **6** after the RRC layer sends alleviation message **456** at **845**.

(74) FIG. **9** illustrates details of example control transactions between entities for informing an upper layer of barring alleviation for multiple access classes responsive to a selection of a different cell. For simplicity, similar transactions as described above with respect to FIGS. **4-7** are not shown explicitly and are generically referenced using identifiers A**1** through F. In this situation, an operation within the transactions identified by C direct the UE RRC layer **404** to stop the common bar timer and inform the UE upper layer **402** of alleviation of the barring of multiple access classes.

(75) At **905** and **910**, operations identified by A**1** and A**2** of FIG. **5** and operations identified by B of FIG. **4** can be performed. During the operations of B, the common bar timer starts as shown by element **435** in FIG. **4**. Prior to the operations identified by C, the UE RRC layer **404** selects a first cell while in a disconnected RRC state (see FIG. **4** element **410**).

(76) At **915**, the UE RRC layer **404** performs a cell selection procedure (e.g., a cell selection or a cell re-selection procedure) that selects a second cell that differs from the first cell. The cell selection procedure can be performed in response to the UE **110** moving to a different location or in response to a PLMN selection request (not shown) received by the UE RRC layer **404** from the UE upper layer **402**.

(77) Responsive to the selection of the second cell, the UE RRC layer **404** stops the common bar timer at **920** and sends the alleviation message **456** to the UE upper layer **402** at **930**. The alleviation message **456** indicates alleviation of the barring of the at least two access classes (e.g., first access class **422** and second access class **444**). If a specific bar timer was started at **905** (as shown in FIG. **5** element **515**), the specific bar timer can also be stopped at **925** in response to the UE RRC layer **404** selecting the second cell. Instead of waiting for the common bar timer to expire as described in FIGS. **4-7**, the operations of FIG. **9** quickly inform the UE upper layer **402** that access classes that were previously barred are alleviated responsive to the UE RRC layer **404** selecting the different cell. Although not explicitly shown, the UE can also perform operations identified by E and F in FIGS. **4** and **6** after the RRC layer sends the alleviation message **456** at **930**.

(78) Generally, the operations and transactions described with respect to FIGS. **4-9** can be combined in a variety of different ways. For example, operations identified by A**1** and A**2**, may or may not occur in different situations. Sometimes, more than two specific bar timers associated with different particular access classes may be setup during A**2** and one or more of these timers may be stopped responsive to the common bar timer expiring (as shown in FIG. **7**) while one or more other timers are allowed to expire independent of the common bar timer (as shown in FIGS. **5** and **6**). As another example, the common bar timer in FIGS. **4-7** or the specific bar timer in FIGS. **5-7** can be stopped responsive to the grant message **832** in FIG. **8** or the cell selection procedure of element **915** in FIG. **9**. In other cases, the third access class **508** described in FIGS. **5** and **7** can be the second access class **444** that is also barred due to the rejection message **432** in FIG. **4**.

(79) In the examples above, the common bar timer is associated with a group of at least two access classes. For situations in which the common bar timer is associated with a single access class, such as the first access class **422**, the operations for informing an upper layer of barring alleviation for multiple access classes as described in FIGS. **5-7** can still occur. In this case, the barring message **442** of FIG. **4** indicates that the first access class **422** is temporarily barred (the second access class **444** is not barred). However, other access classes may be previously barred, such as the third access class **508** as described in FIG. **5**. Alleviation of the first access class **422** and the third access class **508** can therefore occur using a single alleviation message **456** (as described in FIGS. **5** and **7**) or

using multiple alleviation messages **456** (as described in combinations of FIGS. **4** and **6**).

Example Method

(80) FIG. **10** depicts an example method **1000** for informing an upper layer of a UE of barring alleviation for multiple access classes. Method **1000** shows a set of operations (or acts) performed but not necessarily limited to the order or combinations in which the operations are illustrated. Further, any of one or more of the operations may be repeated, combined, reorganized, or linked to provide a wide array of additional and/or alternative methods. In portions of the following discussion, reference may be made to environment **100** of FIG. **1** and entities detailed in FIGS. **2-9**, reference to which is made for example only. The techniques are not limited to performance by one entity or multiple entities operating on one device.

(81) At **1002**, an RRC layer of a UE determines that a first access class is not barred. For example, the UE RRC layer **404** of the UE **110** determines that the first access class **422** is not barred, as shown at **420** in FIG. **4**. This determination can be made as part of the UAC procedure.

(82) At **1004**, the RRC layer sends a first connection request message to a network. The first connection request message requests permission to establish a first wireless connection associated with the first access class. For example, the UE RRC layer **404** sends the connection request message **426** to the RAN **140** to request permission to establish (or re-establish) a first wireless connection associated with the first access class **422**, as shown at **425** in FIG. **4**.

(83) At **1006**, the RRC layer receives a rejection message from the network. The rejection message denies the request to establish the first wireless connection. The set of at least two access classes include the first access class and a second access class. For example, the UE RRC layer **404** receives from the RAN **140** the rejection message **432** using lower layers. The rejection message **432** denies the UE RRC layer **404**'s request to establish the first wireless connection using the first access class, as shown at **430** in FIG. **4**. The rejection message **432** directs the UE RRC layer **404** to determine that the first access class **422** and the second access class **444** are temporarily barred. A wait time **434** included within the rejection message **432** can indicate a duration for which the first access class **422** and the second access class **444** are barred.

(84) At **1008**, responsive to the receiving of the rejection message, the RRC layer provides a first barring message to the upper layer. The barring message indicates barring of at least the first access class and a second access class. For example, responsive to receiving the rejection message **432** at **430** in FIG. **4**, the UE RRC layer **404** provides the UE upper layer **402** the barring message **442** at **440** in FIG. **4**, which indicates barring of at least two access classes. The at least two access classes can include the first access class **422** and the second access class **444**, as shown in FIG. **4**, the first access class **422** and the third access class **508** of FIG. **6**, or the first access class **422**, the second access class **444**, and the third access class **508**.

(85) In other situations, multiple barring messages may be provided from the RRC layer to the upper layer at different times. For example, the barring message **442** can be provided to the UE upper layer **402** with one or more access classes (e.g., the first access class **422** and/or the second access class **444**), as described above, and the barring message **522** can be provided to the UE upper layer **402** with the third access class **508**, as shown at **520** in FIG. **5**.

(86) At **1010**, the RRC layer provides a first alleviation message to the upper layer. The first alleviation message indicates alleviation of the barring of the at least two access classes. For example, the UE RRC layer **404** provides the UE upper layer **402** the alleviation message **456** that indicates barring of the at least two access classes is alleviated, as shown at **455** in FIG. **4**. This enables the UE upper layer **402** to direct the UE RRC layer **404** to send a second connection request message to the RAN **140**, as shown by the UE upper layer **402** sending the second request message **466** to the UE RRC layer **404** at **465** in FIG. **4**. The second connection request message requests permission to establish a second wireless connection associated with one access class of the at least two access classes, such as either the first access class **422**, the second access class **444**, or the third access class **508**.

(87) In general, the access classes included within the alleviation message **456** correspond to at least two of the access classes that are included within the one or more barring messages. Therefore the at least two access classes included within the alleviation message **456** can include the first access class **422**, the second access class **444**, and/or the third access class **508**. Sometimes the access classes in the alleviation message **456** can be associated with different barring messages and different bar timers (e.g., such as the first access class **422** of FIG. **4** and the third access class **508** of FIG. **6**). Sometimes an additional alleviation message can be sent for another access class that is not included within the barring message **442**, such as the alleviation message **622** of FIG. **6**.
(88) In different situations, the UE RRC layer **404** can send the alleviation message **456** in response to the common bar timer expiring as shown in FIGS. **4**, **5**, and **7**, in response to receipt of the grant message **832** from the RAN **140** as shown in FIG. **8**, or in response to a cell selection procedure as shown in element **915** in FIG. **9**.
Conclusion
(89) Although techniques for informing an upper layer of barring alleviation for multiple access classes have been described in language specific to features and/or methods, it is to be understood that the subject of the appended claims is not necessarily limited to the specific features or methods described. Rather, the specific features and methods are disclosed as example implementations of informing an upper layer of barring alleviation for multiple access classes.
(90) Some examples are described below.
(91) Example 1: A user equipment (UE) comprising: a processor and memory system coupled to a radio-frequency transceiver and configured to implement a protocol stack including a radio resource control (RRC) layer and an upper layer; the RRC layer configured to: determine that a first access class is not barred; send, to a network, a first connection request message to request establishment of a first connection associated with the first access class; receive, from the network, a rejection message that denies the request to establish the first connection; responsive to receiving the rejection message, send, to the upper layer, a first barring message that indicates barring of at least the first access class and a second access class; send, to the upper layer, a first alleviation message that indicates alleviation of the barring of the first access class and the second access class; receive a first request message from the upper layer associated with one access class of the first access class and the second access class; and responsive to receiving the first request message, send a second connection request message to the network to request establishment of a second connection associated with the one access class.
(92) Example 2: The UE of example 1, wherein the RRC layer is further configured to: perform a unified access control (UAC) procedure, wherein the first access class is determined to not be barred based on the UAC procedure.
(93) Example 3: The UE of example 1, wherein the rejection message includes a first wait time indicating a duration for which the first access class and the second access class are temporarily barred, further comprising: a common bar timer; wherein the RRC layer is further configured to: start the common bar timer with the first wait time responsive to receiving the rejection message, wherein the first alleviation message is sent responsive to an expiration of the common bar timer.
(94) Example 4: The UE of example 3, wherein the RRC layer is further configured to: prior to sending the first alleviation message, determine whether a specific bar timer associated with the first access class or the second access class is active or inactive, wherein the first alleviation message indicates alleviation of the barring of the first access class or the second access class responsive to a determination that the specific bar timer is inactive.
(95) Example 5: The UE of example 3, further comprising: a specific bar timer; wherein the RRC layer is further configured to: prior to the determination that the first access class is not barred, receive a second request message from the upper layer associated with a third access class that is different from the first access class and the second access class; determine that the third access class is barred; send, to the upper layer, a second barring message that indicates barring of the third

access class; start the specific bar timer with a second wait time that causes the specific bar timer to expire after the common bar timer expires; and responsive to an expiration of the specific bar timer, send, to the upper layer, a second alleviation message that indicates alleviation of the barring of the third access class.

(96) Example 6: The UE of example 5, wherein the third access class and the first access class are associated with a same access category type and have different access category numbers.

(97) Example 7: The UE of example 3, further comprising: a specific bar timer; wherein the RRC layer is further configured to: prior to the determination that the first access class is not barred, receive a second request message from the upper layer associated with the second access class; determine that the second access class is barred; send, to the upper layer, a second barring message that indicates barring of the second access class; start the specific bar timer with a second wait time that causes the specific bar timer to expire after the common bar timer expires; and responsive to the expiration of the common bar timer: stop the specific bar timer; and send, to the upper layer, the first alleviation message.

(98) Example 8: The UE of example 3, further comprising: a specific bar timer; wherein the RRC layer is further configured to: prior to the determination that the first access class is not barred, receive a second request message from the upper layer associated with the second access class; determine that the second access class is barred; send, to the upper layer, a second barring message that indicates barring of the second access class; start the specific bar timer with a second wait time that causes the specific bar timer to expire before the common bar timer expires; and responsive to the expiration of the common bar timer send, to the upper layer, the first alleviation message.

(99) Example 9: The UE of example 1, wherein the RRC layer is further configured to: prior to the determination that the first access class is not barred, select a first cell; after sending the first barring message, perform a cell selection procedure that selects a second cell different from the first cell; and responsive to the selection of the second cell, send, to the upper layer, the first alleviation message.

(100) Example 10: The UE of example 9, wherein the rejection message includes a first wait time indicating a duration for which the first access class and the second access class are temporarily barred, further comprising: a common bar timer; wherein the RRC layer is further configured to: start the common bar timer with the first wait time responsive to receiving the rejection message; and stop the common bar timer responsive to the selection of the second cell.

(101) Example 11: The UE of example 1, wherein the RRC layer is further configured to: inform the upper layer, using the barring message, that at least one exempt access class is not barred; wherein the at least one exempt access class is associated with at least one of the following access category types: a mobile-terminated (MT) access type; or an emergency type.

(102) Example 12: The UE of example 1, wherein the RRC layer is further configured to: determine, after sending the first barring message, that a third access class of the at least one exempt access class is not barred, the third access class being different than the first access class and the second access class; send, to the network, a third connection request message to request establishment of a third connection associated with the third access class; receive, from the network, a grant message that approves the requested establishment of the third connection; and responsive to receiving the grant message, send, to the upper layer, the first alleviation message.

(103) Example 13: The UE of example 12, wherein the rejection message includes a first wait time indicating a duration for which the first access class and the second access class are temporarily barred, further comprising: a common bar timer; wherein the RRC layer is further configured to: start the common bar timer with the first wait time responsive to receiving the rejection message; and stop the common bar timer responsive to receiving the grant message.

(104) Example 14: A method for informing an upper layer of a user equipment (UE) of barring alleviation for multiple access classes, the method comprising the UE performing operations including: determining, at a radio resource control (RRC) layer of the UE, that a first access class is

not barred; sending, by the RRC layer, a first connection request message to a network, the first connection request message requesting permission to establish a first wireless connection associated with the first access class; receiving, at the RRC layer, a rejection message from the network, the rejection message denying the request to establish the first wireless connection; responsive to the receiving of the rejection message, providing, from the RRC layer to the upper layer, a first barring message that indicates barring of at least the first access class and a second access class; and providing, from the RRC layer to the upper layer, a first alleviation message that indicates alleviation of the barring of the first access class and the second access class.

(105) Example 15: The method of example 14, further comprising performing a unified access control procedure at the RRC layer of the UE; and wherein the determining that the first access class is not barred is based on the unified access control procedure.

(106) Example 16: The method of example 14, wherein the rejection message includes a first wait time indicating a duration for which the first access class and the second access class are temporarily barred, and further comprising: responsive to receiving the rejection message, starting, at the RRC layer, a common bar timer with the first wait time, wherein the providing of the first alleviation message is responsive to an expiration of the common bar timer.

(107) Example 17: The method of example 16, further comprising: prior to the determining that the first access class is not barred, receiving, at the RRC layer from the upper layer, a first request message associated with a third access class that is different from the first access class and the second access class; determining, at the RRC layer, that the third access class is barred; providing, from the RRC layer to the upper layer, a second barring message that indicates barring of the third access class; starting a specific bar timer with a second wait time that causes the specific bar timer to expire after the common bar timer expires; and responsive to an expiration of the specific bar timer, providing, from the RRC layer to the upper layer, a second alleviation message that indicates alleviation of the barring of the third access class.

(108) Example 18: The method of example 16, further comprising: prior to the determining that the first access class is not barred, receiving, at the RRC layer from the upper layer, a second request message associated with the second access class; determining, at the RRC layer, that the second access class is barred; providing, from the RRC layer to the upper layer, a second barring message that indicates barring of the second access class; starting a specific bar timer with a second wait time that causes the specific bar timer to expire after the common bar timer expires; and responsive to the expiration of the common bar timer, stopping the specific bar timer.

(109) Example 19: The method of example 14, further comprising: selecting, prior to the determining that the first access class is not barred, a first cell; and performing, after the providing of the first barring message, a cell selection procedure that selects a second cell different than the first cell, wherein the providing the first alleviation message is responsive to the performing of the cell selection procedure.

(110) Example 20: The method of example 14, further comprising: determining, after the sending of the first barring message, that a third access class is not barred at the RRC layer, the third access class being different than the first access class and the second access class; sending a second connection request message from the RRC layer to the network to request establishment of a second wireless connection associated with the third access class; receiving, from the network at the RRC layer, a grant message that approves the second connection request message; and responsive to the receiving of the grant message, providing, from the RRC layer to the upper layer, the first alleviation message.

Claims

1. A method for informing an upper layer of a user equipment of barring alleviation for multiple access classes, the method comprising the user equipment performing operations including:

receiving, at a radio resource control layer from the upper layer, a first request message associated with a fourth access class that is different from a first access class and a second access class; determining, at the radio resource control layer, that the fourth access class is barred; providing, from the radio resource control layer to the upper layer, a second barring message that indicates barring of the fourth access class; starting a first specific bar timer with a first wait time; determining, at the radio resource control layer of the user equipment, that the first access class is not barred; sending, by the radio resource control layer, a first connection request message to a network, the first connection request message requesting permission to establish a first wireless connection associated with the first access class; receiving, at the radio resource control layer, a rejection message from the network, the rejection message denying the request to establish the first wireless connection; responsive to the receiving of the rejection message, providing, from the radio resource control layer to the upper layer, a first barring message that indicates barring of the first access class and the second access class; responsive to receiving the rejection message, activating, at the radio resource control layer, a common bar timer associated with the access classes identified in the first barring message; responsive to the common bar timer becoming inactive, determining whether or not respective specific bar timers associated with the first access class and the second access class are inactive; responsive to determining that the respective specific bar timers associated with the first access class and the second access class are inactive, determining that the barring of the first access class and the second access class is alleviated; providing, from the radio resource control layer to the upper layer, a first alleviation message that indicates alleviation of the barring of the first access class and the second access class; and if the first wait time causes the first specific bar timer to expire after the common bar timer becomes inactive, and responsive to an expiration of the first specific bar timer, providing, from the radio resource control layer to the upper layer, a second alleviation message that indicates alleviation of the barring of the fourth access class.

2. The method of claim 1, wherein: the first barring message indicates barring of the first access class, the second access class, and a third access class; the determining whether or not the respective specific bar timers associated with the first access class and the second access class are inactive comprises determining whether or not the respective specific bar timers associated with the first access class, the second access class, and the third access class are inactive; the determining that the barring of the first access class and the second access class is alleviated comprises determining that the barring of the first access class, the second access class, and the third access class is alleviated responsive to the determining that the respective specific bar timers associated with the first access class, the second access class, and the third access class are inactive; and the first alleviation message indicates alleviation of the barring of the first access class, the second access class, and the third access class.

3. The method of claim 1, further comprising: responsive to the first specific bar timer expiring prior to the common bar timing becoming inactive, determining that the barring of the fourth access class is also alleviated, wherein the first alleviation message indicates alleviation of the barring of the first access class, the second access class, and the fourth access class.

4. The method of claim 1, wherein the fourth access class and the first access class are associated with a same access category type and have different access category numbers.

5. The method of claim 1, further comprising: determining, after the sending of the first barring message, that a fifth access class is not barred at the radio resource control layer, the fifth access class being different than the first access class and the second access class; sending a second connection request message from the radio resource control layer to the network to request establishment of a second wireless connection associated with the fifth access class; receiving, from the network at the radio resource control layer, a grant message that approves the second connection request message; and responsive to the receiving of the grant message, stopping the common bar timer.

6. The method of claim 1, further comprising: prior to the determining that the first access class is not barred, receiving, at the radio resource control layer from the upper layer, a second request message associated with the second access class; determining, at the radio resource control layer, that the second access class is barred; providing, from the radio resource control layer to the upper layer, a third barring message that indicates barring of the second access class; starting a second specific bar timer with a second wait time that causes the second specific bar timer to expire after the common bar timer expires; and responsive to the common bar timer becoming inactive, stopping the second specific bar timer prior to the determining whether or not the respective specific bar timers associated with the first access class and the second access class are inactive.
7. The method of claim 1, further comprising: prior to the determining that the first access class is not barred, receiving, at the radio resource control layer from the upper layer, a second request message associated with the second access class; determining, at the radio resource control layer, that the second access class is barred; providing, from the radio resource control layer to the upper layer, a third barring message that indicates barring of the second access class; and starting a second specific bar timer with a second wait time that causes the second specific bar timer to expire before the common bar timer expires.
8. The method of claim 1, further comprising: selecting, prior to the determining that the first access class is not barred, a first cell; performing, after the providing the first barring message, a cell selection procedure that selects a second cell different than the first cell; and responsive to selecting the second cell, stopping the common bar timer.
9. The method of claim 1, wherein: the first barring message indicates that at least one exempt access class is not barred; and the at least one exempt access class is associated with at least one of the following access category types: a mobile-terminated access type; or an emergency type.
10. The method of claim 1, further comprising: performing a unified access control procedure at the radio resource control layer of the user equipment, wherein the determining that the first access class is not barred is based on the unified access control procedure.
11. The method of claim 1, wherein: the rejection message includes a third wait time indicating a duration for which the first access class and the second access class are temporarily barred; and the activating the common bar timer comprises starting the common bar timer with the third wait time.
12. The method of claim 1, wherein the common bar timer becomes inactive based on the common bar timer expiring or the common bar timer stopping.
13. The method of claim 1, wherein the first access class and the second access class are non-exempt access classes.
14. A user equipment comprising: a radio-frequency transceiver; and a processor and memory system configured to: receive, at a radio resource control layer from an upper layer, a first request message associated with a fourth access class that is different from a first access class and a second access class; determine, at the radio resource control layer, that the fourth access class is barred; provide, from the radio resource control layer to the upper layer, a second barring message that indicates barring of the fourth access class; start a first specific bar timer with a first wait time; determine, at the radio resource control layer of the user equipment, that the first access class is not barred; send, by the radio resource control layer, a first connection request message to a network using the radio-frequency transceiver, the first connection request message requesting permission to establish a first wireless connection associated with the first access class; receive, at the radio resource control layer, a rejection message from the network using the radio-frequency transceiver, the rejection message denying the request to establish the first wireless connection; responsive to the receiving of the rejection message, provide, from the radio resource control layer to an upper layer, a first barring message that indicates barring of the first access class and the second access class; responsive to receiving the rejection message, activate, at the radio resource control layer, a common bar timer associated with the access classes identified in the first barring message; responsive to the common bar timer becoming inactive, determine whether or not respective

specific bar timers associated with the first access class and the second access class are inactive; responsive to determining that the respective specific bar timers associated with the first access class and the second access class are inactive, determine that the barring of the first access class and the second access class is alleviated; provide, from the radio resource control layer to the upper layer, a first alleviation message that indicates alleviation of the barring of the first access class and the second access class; and if the first wait time causes the first specific bar timer to expire after the common bar timer becomes inactive, and responsive to an expiration of the first specific bar timer, provide, from the radio resource control layer to the upper layer, a second alleviation message that indicates alleviation of the barring of the fourth access class.
15. A non-transitory processor-readable medium having instructions stored thereon that, when executed by a processor, cause the processor to: receive, at a radio resource control layer from an upper layer, a first request message associated with a fourth access class that is different from a first access class and a second access class; determine, at the radio resource control layer, that the fourth access class is barred; provide, from the radio resource control layer to the upper layer, a second barring message that indicates barring of the fourth access class; start a first specific bar timer with a first wait time; determine, at the radio resource control layer of a user equipment, that the first access class is not barred; send, by the radio resource control layer, a first connection request message to a network, the first connection request message requesting permission to establish a first wireless connection associated with the first access class; receive, at the radio resource control layer, a rejection message from the network, the rejection message denying the request to establish the first wireless connection; responsive to the receiving of the rejection message, provide, from the radio resource control layer to an upper layer, a first barring message that indicates barring of the first access class and the second access class; responsive to receiving the rejection message, activate, at the radio resource control layer, a common bar timer associated with the access classes identified in the first barring message; responsive to the common bar timer becoming inactive, determine whether or not respective specific bar timers associated with the first access class and the second access class are inactive; responsive to determining that the respective specific bar timers associated with the first access class and the second access class are inactive, determine that the barring of the first access class and the second access class is alleviated; provide, from the radio resource control layer to the upper layer, a first alleviation message that indicates alleviation of the barring of the first access class and the second access class; and if the first wait time causes the first specific bar timer to expire after the common bar timer becomes inactive, and responsive to an expiration of the first specific bar timer, provide, from the radio resource control layer to the upper layer, a second alleviation message that indicates alleviation of the barring of the fourth access class.
16. The user equipment of claim 14, wherein: the first barring message indicates barring of the first access class, the second access class, and a third access class; the processor and memory system are further configured to: determine whether or not the respective specific bar timers associated with the first access class, the second access class, and the third access class are inactive; and determine that the barring of the first access class, the second access class, and the third access class is alleviated responsive to the determining that the respective specific bar timers associated with the first access class, the second access class, and the third access class are inactive; and the first alleviation message indicates alleviation of the barring of the first access class, the second access class, and the third access class.
17. The user equipment of claim 14, wherein: the processor and memory system are further configured to: responsive to the first specific bar timer expiring prior to the common bar timing becoming inactive, determine that the barring of the fourth access class is also alleviated; and the first alleviation message indicates alleviation of the barring of the first access class, the second access class, and the fourth access class.
18. The user equipment of claim 14, wherein the first access class and the second access class are

non-exempt access classes.
19. The user equipment of claim 14, wherein the fourth access class and the first access class are associated with a same access category type and have different access category numbers.
20. The user equipment of claim 14, wherein: the first barring message indicates that at least one exempt access class is not barred; and the at least one exempt access class is associated with at least one of the following access category types: a mobile-terminated access type; or an emergency type.